# Partial Differential Equations (PDE)
# Assignment 6

[Laplace equation and Green function]

(a) Show that in 3D the Green's function for a half-space $D = \{z > 0\}$ is

$$G(\mathbf{x}, \mathbf{x}_0) = -\frac{1}{4\pi|\mathbf{x} - \mathbf{x}_0|} + \frac{1}{4\pi|\mathbf{x} - \mathbf{x^*}_0|}$$

where $\mathbf{x} = (x, y, z)$ and $\mathbf{x}^* = (x, y, -z)$.

(b) Show that the solution of Laplace equation

$$\begin{cases} \triangle u = 0, & \text{in } D \\ u = h, & \text{on } \partial D \end{cases} \tag{1}$$

is represented by

$$u(\mathbf{x}_0) = \iint_{\partial D} \frac{1}{2\pi} \frac{z_0}{|\mathbf{x} - \mathbf{x^*}_0|^3} h(\mathbf{x}) dS$$

[Hint]

$$u(\mathbf{x}_0) = \iint_{\partial D} \frac{\partial G(\mathbf{x}, \mathbf{x}_0)}{\partial n} h(\mathbf{x}) dS$$

solves Eq. (1).

(c) For two dimensional case, show that the Green's function for the half-plane $D = \{y > 0\}$ is

$$G(\mathbf{x}, \mathbf{x}_0) = \frac{1}{2\pi} \log|\mathbf{x} - \mathbf{x}_0| - \frac{1}{2\pi} \log|\mathbf{x} - \mathbf{x^*}_0|$$

where $\mathbf{x} = (x, y)$ and $\mathbf{x}^* = (x, -y)$. Represent the solution of Eq. (1).